

ONGC Summer Internship Plan: Chatbot Development using LLM (Ollama) & OpenWebUI (One-Month Duration)

Objective:

To introduce interns to Large Language Model (LLM)-based chatbot development using open-source tools like Ollama and OpenWebUI, providing hands-on experience in building, testing, and deploying a functional chatbot aligned to stated requirements.

Week 1: Foundation and Tools Setup

Goal: Understand LLMs, chatbot architecture, and set up environment.

Day 1-2: Induction & Introduction

- Welcome by Training Coordinator.
- Brief on ONGC's digital transformation and AI roadmap.
- Introduction to Chatbots, LLMs (like LLaMA), and their relevance.

Day 3-5: Tool Setup and Exploration

- Install and configure Ollama LLM locally (or via Docker).
- Setup OpenWebUI interface for chatbot front-end.
- Basic prompt engineering and model testing using UI.

Deliverable: Working chatbot interface connected to Ollama with basic Q&A functionality.

Week 2: Backend Development and Data Integration

Goal: Enable chatbot to use structured knowledge and specific content.

Day 6-7: Custom Prompt Engineering & Workflow Design

- Prompt templates & FAQs.
- Workflow: prompt > model > response > UI.

Day 8-9: Data Source Integration

- Load PDFs, DOCs, or text files into vector database (e.g., ChromaDB).
- Connect vector store to Ollama using RAG (Retrieval-Augmented Generation).
- Use LangChain or similar if needed for data pipeline.

Day 10: Mini Project – Part 1

- Interns work individually to build chatbot on selected documents.

Deliverable: Chatbot that answers queries from Knowledge Base of PDFs.

Week 3: UI Enhancement and Advanced Features

Goal: Enhance usability and implement multi-domain support.

Day 11-12: OpenWebUI Customization

- Manage frontend.
- Add dropdown to switch between knowledge base domains.

Day 13-14: Logging, Feedback, and Analytics

- Enable logging of user queries and responses.
- Add feedback button for each answer.
- Analyze response accuracy and model limitations.

Day 15: Mini Project – Part 2

- Add module using LLM + document store.

Deliverable: Multi-domain chatbot with feedback and basic analytics.

Week 4: Final Project & Deployment Plan

Goal: Build final chatbot prototype and demonstrate value.

Day 16-18: Final Project Work

- Finalize chatbot with modules.
- Polish UI, improve accuracy, optimize prompt design.

Day 19: Presentation Preparation

- Create technical documentation.
- Prepare demonstration for Mentors.

Day 20: Final Presentation & Evaluation

- Individual presents to senior officers.
- Live demo of chatbot.
- Feedback.

Deliverable: Fully functional chatbot demo + documentation + deployment plan.

Support Throughout Internship:

- **Mentorship, Guidance, Tools & Learning Resources**
 - *Build Your Own Chatbot with Langchain, Ollama & LLAMA 3.2 | Local LLM Tutorial*
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